

Problem 1. Translate these statements into logic using the provided propositions.

$v \equiv$ The message is scanned for viruses.

$s \equiv$ The message is marked as spam.

$k \equiv$ The message was sent from a known system.

$b \equiv$ The sender is blocked.

- a) The message is scanned for viruses or marked as spam.
- b) A message from a blocked sender goes scanned for viruses.
- c) The sender is blocked and the message is scanned for viruses only if the message was sent from an unknown system.
- d) If the message is sent from an unknown system, either the sender is blocked or the message is marked for spam. In both cases, it is scanned for viruses.

Answer.

Problem 2. Prove the following by using **logical equivalences** (i.e., using direct manipulation). For each step, state which single logical equivalence was used.

- a) $\neg(p \rightarrow \neg q) \equiv (p \wedge q)$
- b) $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Answer.

Problem 3. Use the truth tables to:

- a) Show that $p \vee (p \wedge q) \equiv p$
- b) Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent

Answer.

Problem 4. Let $M(x) \equiv x \text{ visits the Carnegie Museum}$ and $A(x) \equiv x \text{ is an art student}$. Let the domain of each be “all students at Pitt”.

- a) Translate the following statement into English.

$$\forall x [M(x) \rightarrow A(x)]$$

- b) Translate the following into predicate logic.
“Not all art students have visited the Carnegie Museum”.

- c) Explain why the statement “Some art students visit the Carnegie Museum”
cannot be translated as:

$$\exists x [A(x) \rightarrow M(x)]$$

Answer.

Problem 5. Explain the following logical paradoxes. That is, explain why each one cannot be assigned a specific value.

- a) “The next statement is true. The previous statement is false.”
- b) “If this sentence is true, then all statements are true.”
- c) Define two propositional variables as $p \equiv q$ is false, and $q \equiv p$ is false.

Answer.